# Your grade: 98.33%

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Instructions

**1.** Which of the following is a common method for securing AI models against adversarial attacks? 1 / 1 point

- (•) Adversarial training
- ( ) Hyperparameter tuning
- ( ) Data augmentation
- ( ) Regularization

**Nice work**

Correct! Adversarial training is a technique where models are trained on adversarial examples to improve their robustness.

**2.** What is the primary goal of federated learning in the context of AI? 1 / 1 point

- ( ) To ensure data privacy by anonymizing the training data
- (•) To train a central model using data from multiple sources without transferring the data to a central server
- ( ) To enhance model performance by hyperparameter optimization
- ( ) To split a model into smaller sub-models and train them separately

**Nice work**

Correct! Federated learning enables training a model across multiple decentralized devices or servers holding local data samples without exchanging them.

**3.** Select the techniques that can be used to improve the transparency of AI models. (Select all that apply) 1 / 1 point

- [ ] Data encryption
- [x] Model interpretability tools

**Nice work**

Correct! Model interpretability tools help in understanding and explaining how models make decisions.

- [x] Feature importance analysis

**Nice work**

Correct! Analyzing feature importance helps in understanding which features most influence the model's decisions.

- [ ] Black-box testing
- [x] Surrogate models

**Nice work**

Correct! Surrogate models can mimic black-box models and provide insights into their decision-making process.

**4.** Which metric would you use to evaluate the overall performance of a binary classification model? 1 / 1 point

- ( ) Recall
- ( ) Specificity
- (•) AUC (Area Under the Curve)
- ( ) Precision

**Nice work**

Correct! AUC evaluates the overall performance of a binary classification model by measuring the area under the ROC curve.

**5.** Which steps are involved in the analytic development process for training AI models? (Select all that apply) 0.8 / 1 point

- [ ] User interface design
- [x] Data collection

**Nice work**

Correct! Data collection is one of the initial steps in the analytic development process.

- [ ] Model deployment
- [ ] Market analysis
- [x] Hyperparameter tuning

**Nice work**

Correct! Hyperparameter tuning is an essential step to optimize the model's performance.

- [x] Feature engineering

**Nice work**

Correct! Feature engineering is an important step where raw data is transformed into features suitable for modeling.

**You didn't select all the correct answers**

**6.** What is the primary use of the ROC curve in evaluating AI models? 1 / 1 point

- ( ) To measure the accuracy of the model
- ( ) To identify the most important features in the model
- ( ) To determine the optimal hyperparameters for the model
- (•) To visualize the trade-off between true positive rate and false positive rate

**Nice work**

Correct! The ROC curve is used to visualize the trade-off between the true positive rate and false positive rate for different thresholds.

**7.** What is a primary reason for incorporating AI into network sensors for cybersecurity? 1 / 1 point

- (•) To enhance real-time threat detection
- ( ) To simplify the network architecture
- ( ) To reduce the need for human analysts entirely
- ( ) To minimize network bandwidth usage

**Nice work**

Correct! Incorporating AI into network sensors helps in enhancing real-time threat detection.

**8.** Which of the following is a key consideration in the development process of AI models for cybersecurity? 1 / 1 point

- (•) Ensuring data quality and diversity
- ( ) Using only labeled data for training
- ( ) Minimizing computational costs at all costs
- ( ) Focusing solely on the accuracy of the model

**Nice work**

Correct! Ensuring data quality and diversity is critical for developing robust AI models in cybersecurity.

**9.** Which of the following challenges are associated with using AI for triaging cybersecurity alerts? (Select all that apply) 1 / 1 point

- [x] Data privacy concerns

**Nice work**

Correct! Data privacy is a major concern when using AI in cybersecurity.

- [ ] Ease of implementation
- [x] High volume of false positives

**Nice work**

Correct! One of the main challenges with AI in triaging alerts is the high volume of false positives.

- [x] Limited scalability of AI models

**Nice work**

Correct! Scaling AI models to handle large volumes of data is a significant challenge.

**10.** What are important data quality considerations when developing AI models for cybersecurity? (Select all that apply) 1 / 1 point

- [x] Timeliness of data

**Nice work**

Correct! Timely data is crucial for real-time threat detection and response.

- [x] Data diversity

**Nice work**

Correct! Data diversity ensures that the model can generalize well across different scenarios.

- [ ] Exclusively using synthetic data
- [x] Data labeling accuracy

**Nice work**

Correct! Accurate labeling is essential for training effective AI models.